

Arduino, available as open source design, was obviously picked up by a lot of companies and the prices have shot down to new lows. Several new types were developed with specific usage in mind. “Arduino” has been to “Maker Tech”, what “Android” was to the world of mobile OS.

The community grew larger and the available resources increased every day. Eventually, it gave rise to interchangeable add-on modules known as shields. They are printed circuit expansion boards and are perfectly compatible with the I/O pins provided in Arduino. They can provide the added capabilities of motor controls, GPS, breadboarding, etc.

Conventionally, Arduino boards have used AtMel 8-, 16-, or 32-bit microcontrollers, specifically the AtMel megaAVR series of chips, and it is only recently that other microcontrollers are being used. The first board labelled Arduino, the Serial Arduino used a 16MHz ATmega8 and was about 80mm by 50mm in dimensions.

Currently, according to the official Arduino website, the Arduino boards can be classified in the following manner:

- ♦ ENTRY LEVEL: Arduino Uno, Arduino 101, Arduino Pro
- ♦ ENHANCED FEATURES: Arduino Mega, Arduino Zero, Arduino Due
- ♦ INTERNET OF THINGS: Arduino Yun
- ♦ WEARABLES: Arduino Gemma, Lilipad Arduino (USB, Main Board, Simple, Simple Snap variants)

To go into the details of each one of them would make this chapter too lengthy. How about we have a better look at few of the popular ones?

Arduino Uno (Current version: Rev 3)

One of the entry level boards at about 1400INR at the time of this chapter being written, some Indian ecommerce sites have also listed variants at 600INR. This is the most used and best documented board in the entire Arduino family and with 16MHz clock speed, 32KB flash memory and 2KB SRAM, this board is more than sufficient for any beginner’s experiments. And even if you break it, the ATmega328P can be replaced from its slot for quite cheap. It can even be powered via USB.

Although, if your project is more complex than average, and incorporates displays with texts and likewise, you might need something with more SRAM than 2KB.

Arduino Leonardo

Although it has been officially retired by Arduino, you will still find some